
CIS 678 - Machine Learning

– MAP and Bayesian Estimation –

— Instructor: Yong Zhuang —

yong.zhuang@gvsu.edu

Last Class

- Joint and conditional probability
- Learning classifiers through $P(Y | X)$
- Probability vs. likelihood
- Maximum Likelihood Estimation (MLE)
- MLE for Bernoulli variables

Agenda

- From MLE to MAP
- Prior Knowledge and Bayes' Rule
- Beta Prior Distribution
- MAP for Bernoulli Variables
- Conjugate Priors
- Multinomial-Dirichlet Extension
- Parameter Estimation for Gaussian Variables

Estimating the Bias of a Coin

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?

Each flip yields a Boolean value for X , $X \sim \text{Bernoulli}(\theta)$

Bernoulli Random Variable: $P(X=1) = \theta$; $P(X=0) = 1-\theta$



$X=1$

$X=0$

We flip it repeatedly, observing the outcome:

- It turns Heads (i.e. $X=1$) α_H times
- It turns Tails (i.e. $X=0$) α_T times

How can we estimate the probability of heads $\theta = P(X=1)$?



Maximum A posteriori Probability(MAP)

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?



X=1

X=0

Maximum A posteriori Probability

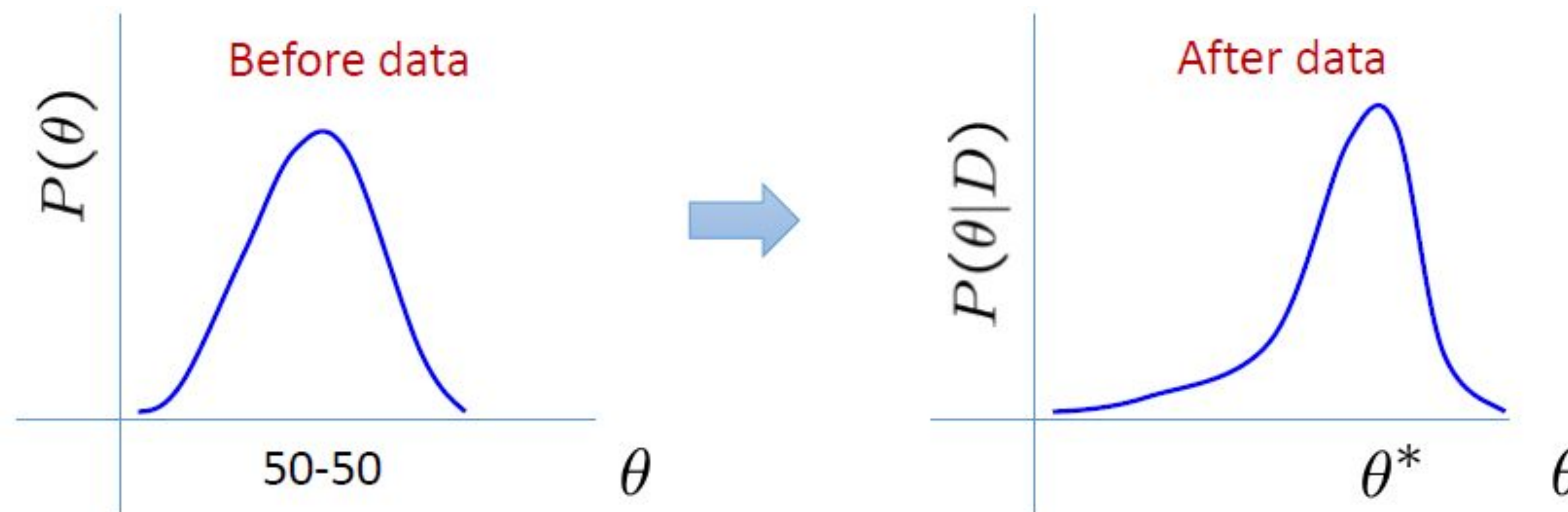
Choose parameter θ that maximizes the posterior probability $P(\theta \mid \text{data})$.

$$\hat{\theta}_{\text{MAP}} = \frac{\alpha_H + \# \text{prior-added heads}}{(\alpha_T + \# \text{prior-added tails}) + (\alpha_H + \# \text{prior-added heads})}$$

What if we have prior knowledge?

Prior Knowledge: E.g., I know that the coin is “close” to 50 - 50.

MAP estimate: we should choose the value of Theta that is most probable, given the observed data D and our prior assumptions summarized by $P(\theta)$.



Bayesian Learning

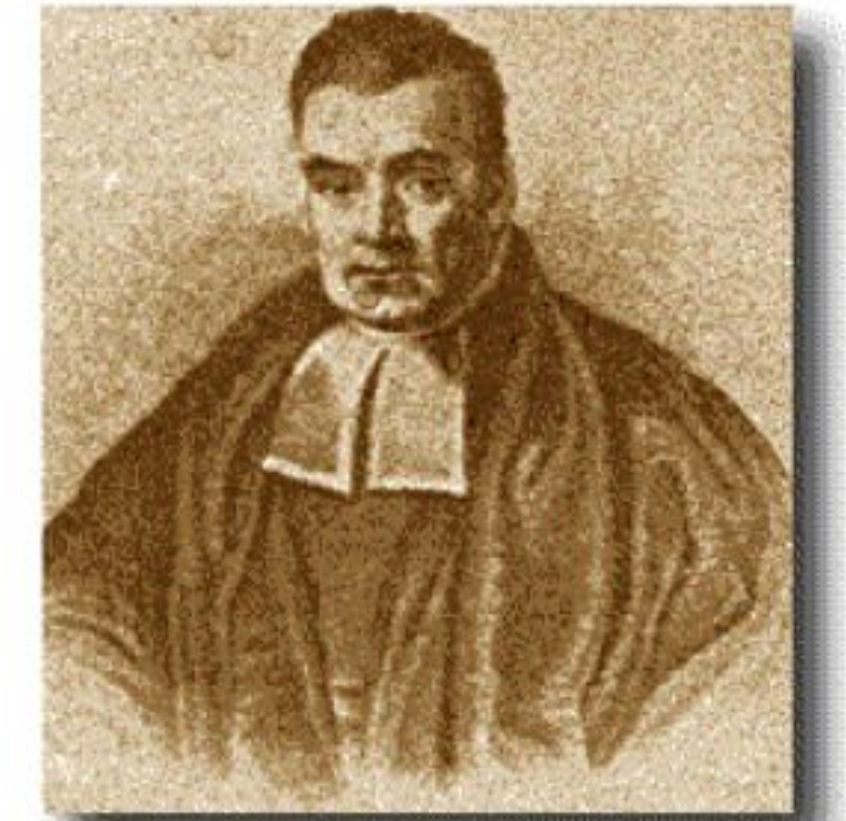
Use Bayes Rule

$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$



$$\underline{P(\theta | D)} \propto \underline{P(D | \theta)} \cdot \underline{P(\theta)}$$

posterior **likelihood** **prior**



Bayes, Thomas (1763) An essay towards solving a problem in the doctrine of chances. *Philosophical Transactions of the Royal Society of London*, 53:370-418

MAP estimate: choose parameter $\hat{\theta}$ that maximizes the posterior probability $P(\hat{\theta} | \text{data})$, i.e., it chooses the value that is most probable given observed data and prior belief.

Maximum A posteriori Probability(MAP)

Problem: Assume we can flip a coin with bias θ several times. Estimate the probability that it turns out heads when we flip it?



X=1



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Maximum A posteriori Probability

Choose parameter θ that maximizes the posterior probability $P(\theta \mid \text{data})$, i.e., it chooses the value that is most probable given observed data and prior belief.

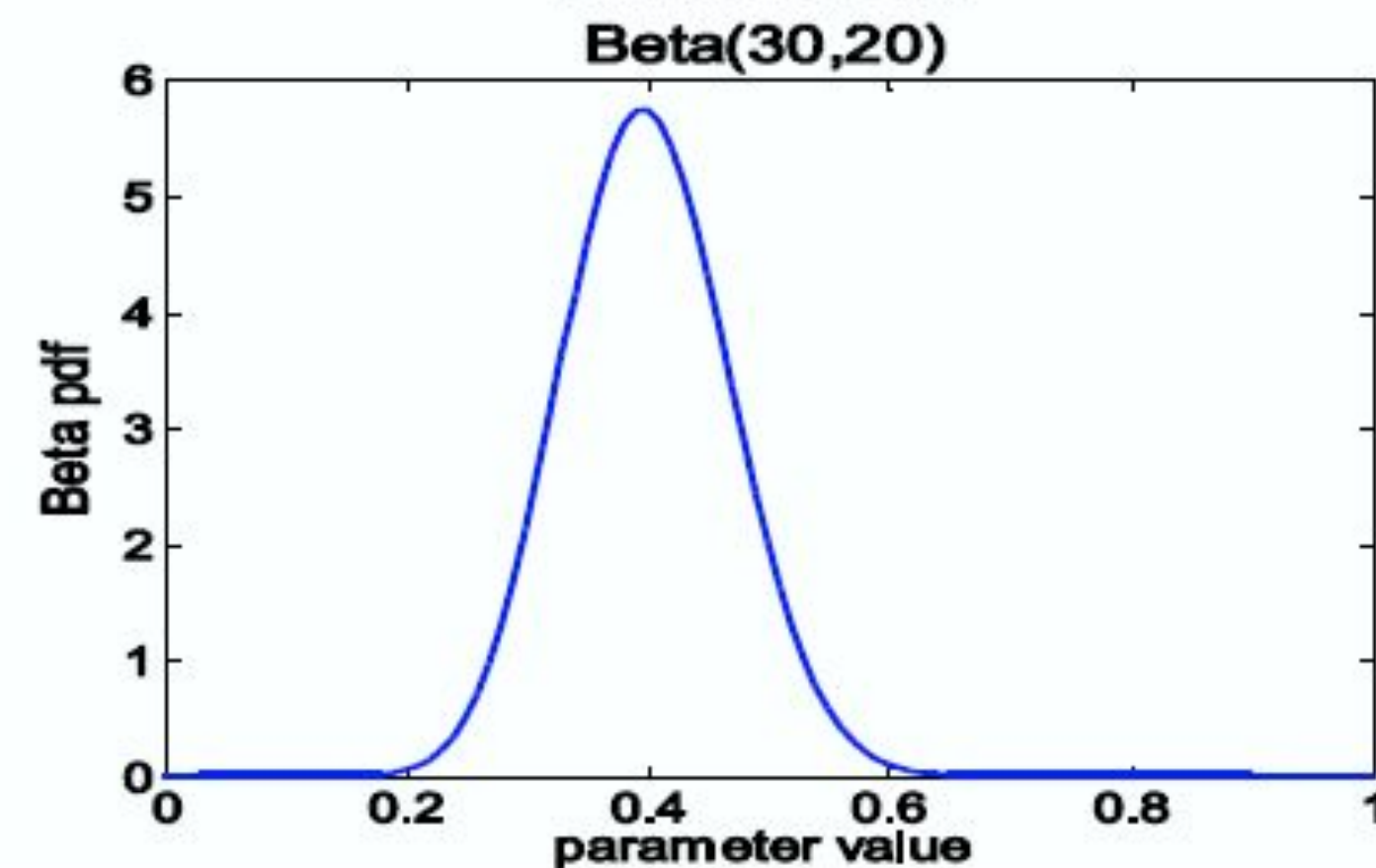
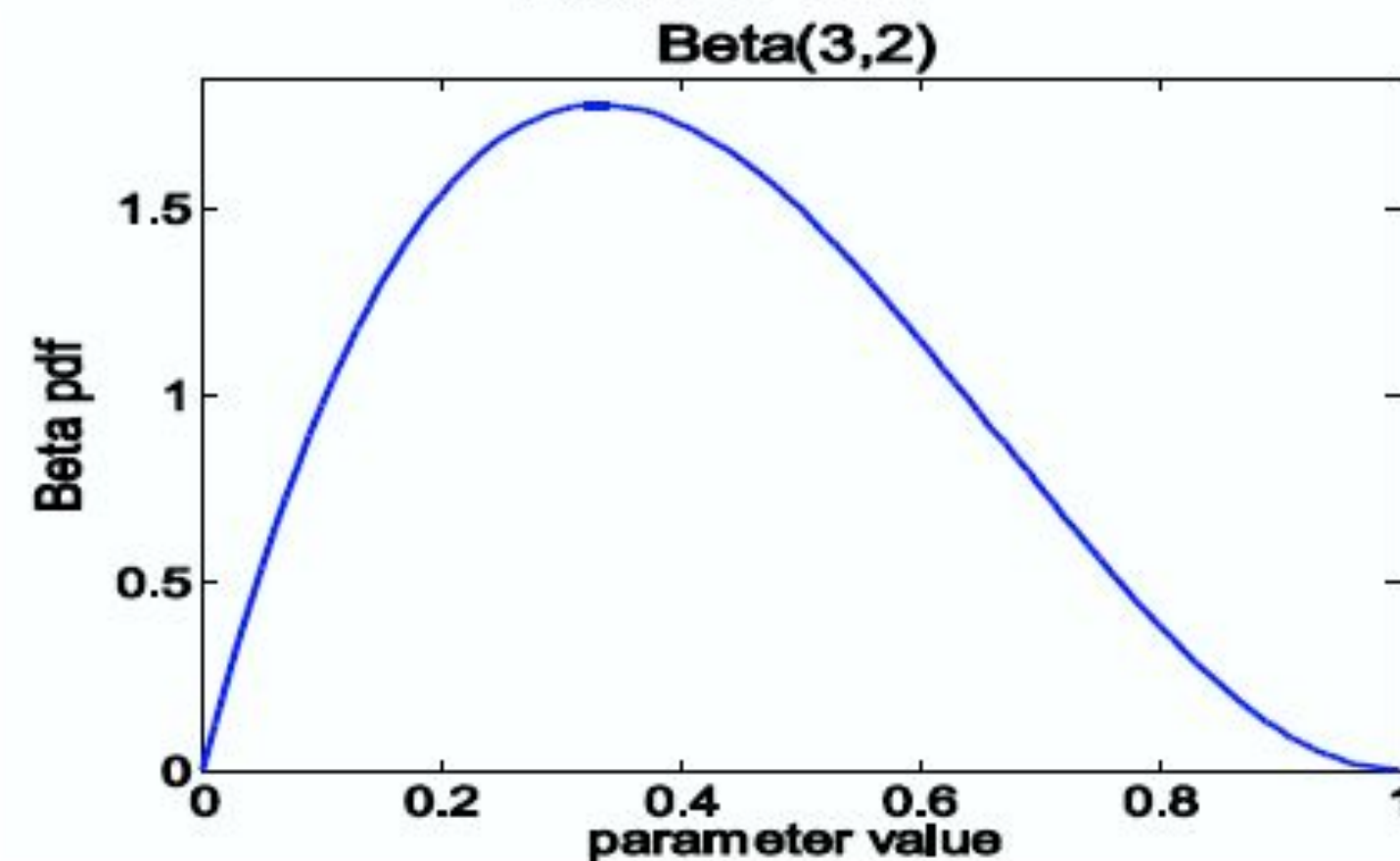
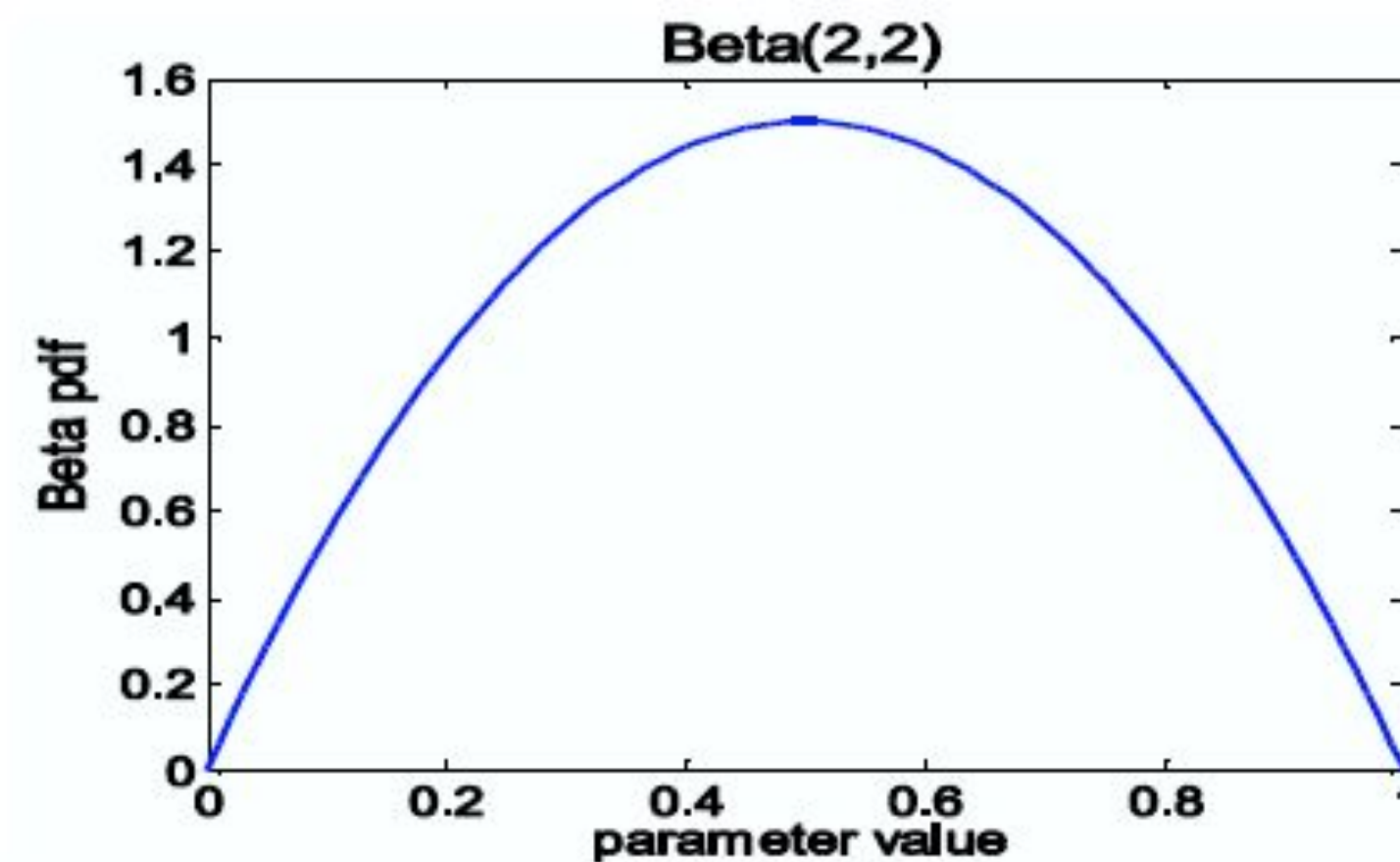
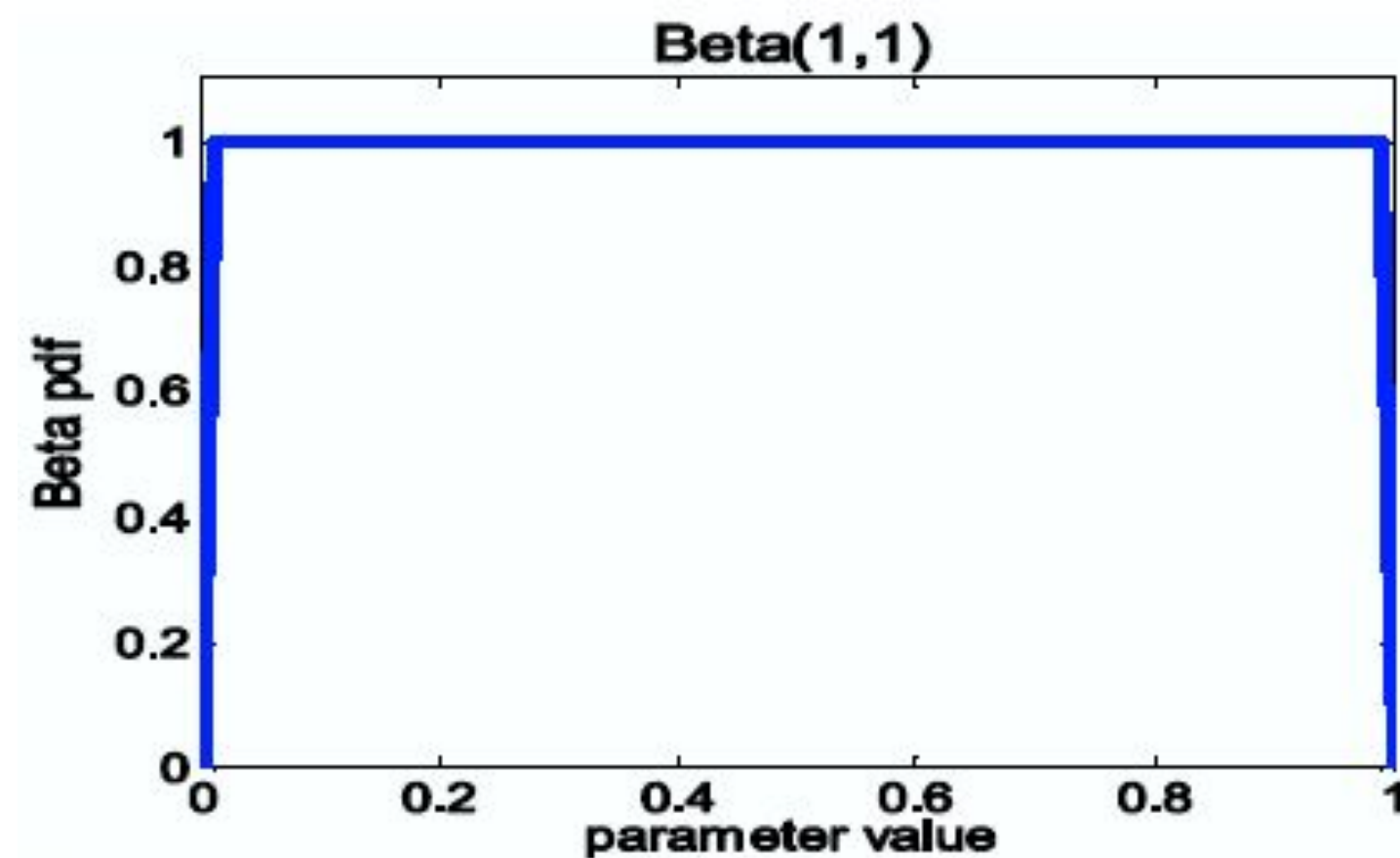
$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} P(\theta \mid D) = \arg \max_{\theta} P(D \mid \theta) P(\theta)$$



- As $n \rightarrow \infty$, prior is forgotten
- For small sample sizes, prior is important

Beta Prior Distribution

Assume $\theta \sim \text{Beta}(\beta_H, \beta_T)$ i.e.,
$$P(\theta) = \frac{\theta^{\beta_H-1}(1-\theta)^{\beta_T-1}}{B(\beta_H, \beta_T)}$$



More concentrated as values of β_H, β_T increase

MAP Estimate for Bernoulli Variables with Beta Prior Distribution

Assume $\theta \sim \text{Beta}(\beta_H, \beta_T)$ **i.e.,** $P(\theta) = \frac{\theta^{\beta_H-1}(1-\theta)^{\beta_T-1}}{B(\beta_H, \beta_T)}$

Likelihood: $P(D \mid \theta) = \theta^{\alpha_H}(1-\theta)^{\alpha_T}$

Posterior: $P(\theta \mid D) \propto P(D \mid \theta)P(\theta)$
 $\propto \theta^{\alpha_H+\beta_H-1}(1-\theta)^{\alpha_T+\beta_T-1}$

Like MLE, but adding β_H-1 prior-added heads and β_T-1 prior-added tails.

$$\hat{\theta}_{\text{MAP}} = \frac{\alpha_H + \beta_H - 1}{(\alpha_T + \beta_T - 1) + (\alpha_H + \beta_H - 1)}$$

Note: as we get more samples, the effect of the prior is reduced.

Conjugate Priors

Likelihood:

Prior:

Posterior:



Conjugate Prior: $P(\theta)$ is the conjugate prior for the likelihood $P(D|\theta)$ if the forms of $P(\theta)$ and $P(\theta|D)$ are the same.

Important: A conjugate prior is not required for MAP.

Conjugacy is useful because it makes the posterior easy to derive:

Prior and posterior belong to the same distribution family.

MAP Estimate for Bernoulli Variables with Beta Prior Distribution

Assume $\theta \sim \text{Beta}(\beta_H, \beta_T)$ **i.e.,** $P(\theta) = \frac{\theta^{\beta_H-1}(1-\theta)^{\beta_T-1}}{B(\beta_H, \beta_T)}$

Likelihood: $P(D \mid \theta) = \theta^{\alpha_H}(1-\theta)^{\alpha_T}$

Posterior: $P(\theta \mid D) \propto P(D \mid \theta)P(\theta)$
 $\propto \theta^{\alpha_H+\beta_H-1}(1-\theta)^{\alpha_T+\beta_T-1} \sim \text{Beta}(\alpha_H + \beta_H, \alpha_T + \beta_T)$

Practice: MAP Estimation for a Coin

Suppose we observe three coin flips:

$$D = \{H, H, T\}$$

so

$$\alpha_H = 2, \quad \alpha_T = 1.$$

Before observing the data, assume

$$\theta \sim \text{Beta}(4, 4).$$

1. What is the posterior distribution $P(\theta \mid D)$?
2. What is the MAP estimate $\hat{\theta}_{\text{MAP}}$?
3. What is the MLE estimate $\hat{\theta}_{\text{MLE}}$?
4. Why are the MAP and MLE estimates different?



MAP Estimate for Dice Rolling with Dirichlet Prior Distribution

Dice Roll Problem: 6 outcomes instead of 2.

Likelihood is $\sim \text{Multinomial}(\theta_1, \dots, \theta_k)$ $P(D | \theta) = \theta_1^{\alpha_1} \theta_2^{\alpha_2} \dots \theta_k^{\alpha_k}$

Prior is Dirichlet distribution, $\theta \sim \text{Dirichlet}(\beta_1, \beta_2, \dots, \beta_k)$

$$P(\theta) = \frac{\prod_{i=1}^k \theta_i^{\beta_i-1}}{B(\beta_1, \beta_2, \dots, \beta_k)}$$

Then **Posterior:** $P(\theta | D) \propto P(D | \theta)P(\theta) \propto \text{Dirichlet}(\alpha_1 + \beta_1, \dots, \alpha_k + \beta_k)$

For Multinomial, conjugate prior is Dirichlet

Summary: Estimating Probabilities

Maximum Likelihood Estimation

Choose parameter $\hat{\theta}$ that maximizes likelihood of observed data $P(\text{data}|\hat{\theta})$

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} P(D | \theta) = \frac{\alpha_H}{\alpha_T + \alpha_H}$$

Maximum A posteriori Probability

Choose parameter θ that maximizes the posterior probability $P(\theta | \text{data})$, i.e., it chooses the value that is most probable given observed data and prior belief.

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} P(\theta | D) = \arg \max_{\theta} P(D | \theta)P(\theta)$$

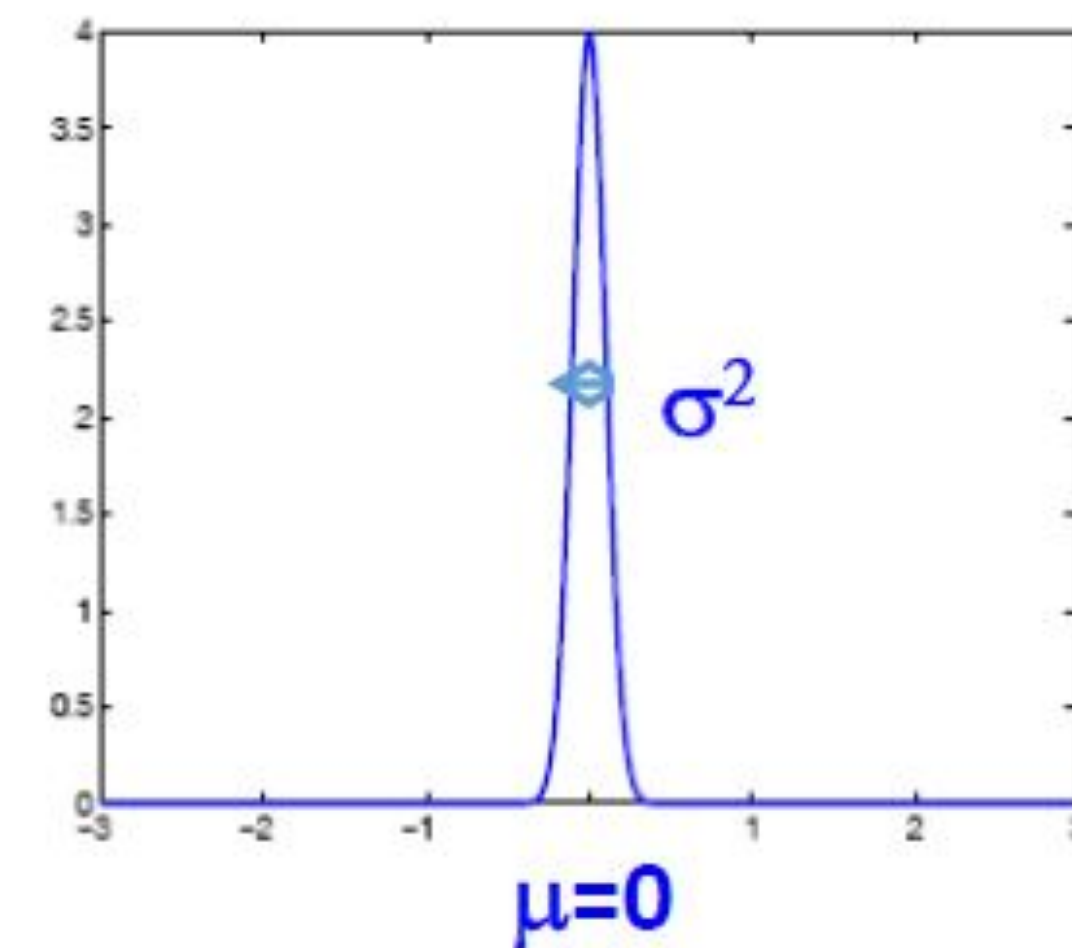
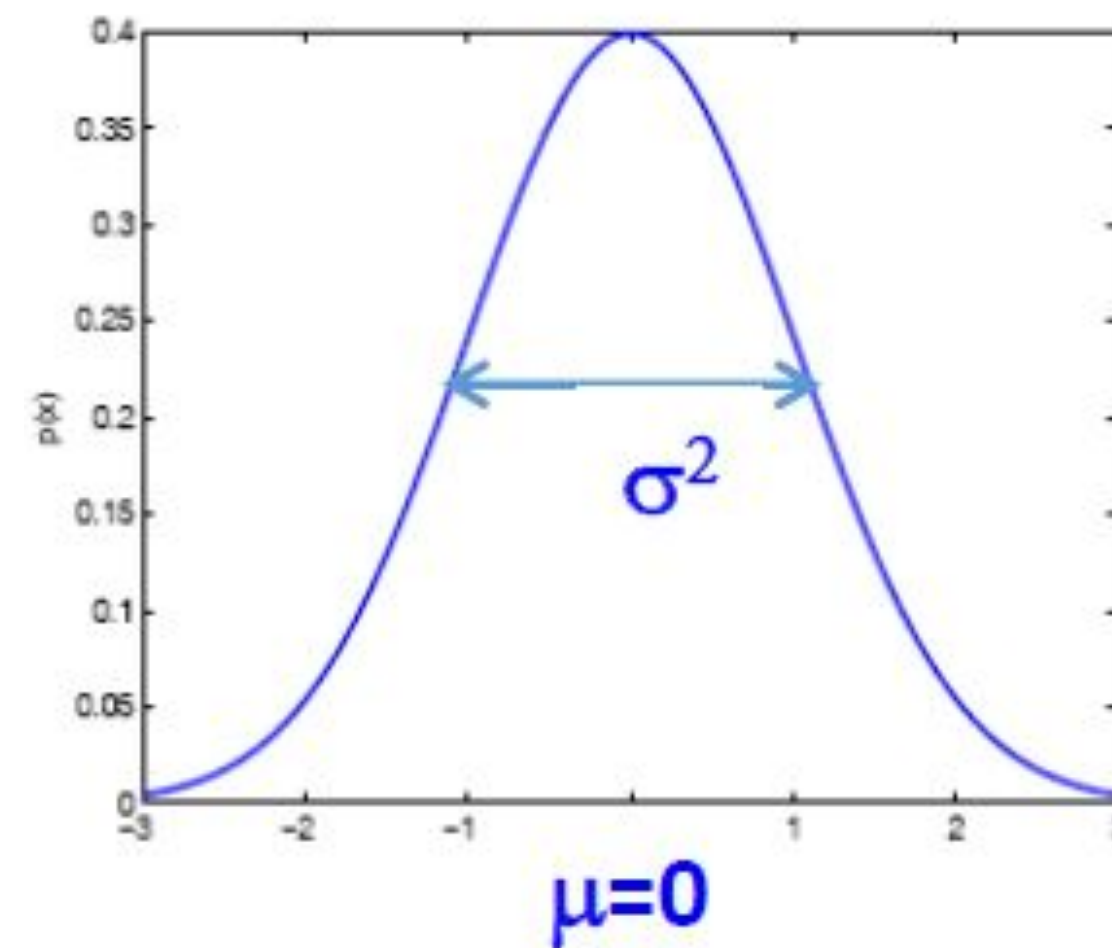
- As $n \rightarrow \infty$, prior is forgotten
- For small sample sizes, prior is important

What About Continuous Random Variables?

Gaussian Random Variable

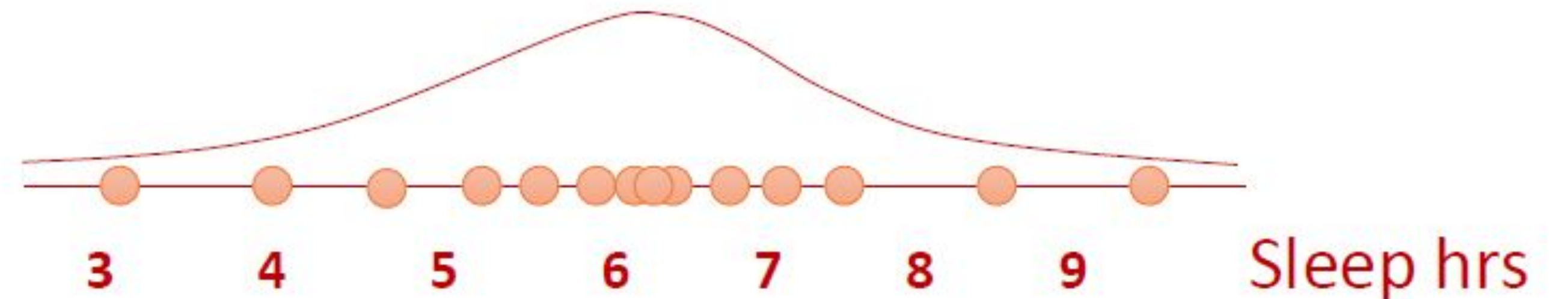
$X \sim \mathcal{N}(\mu, \sigma)$, then

$$p(x \mid \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



What About Continuous Random Variables?

Observed data D:



Parameters: μ mean, σ^2 variance

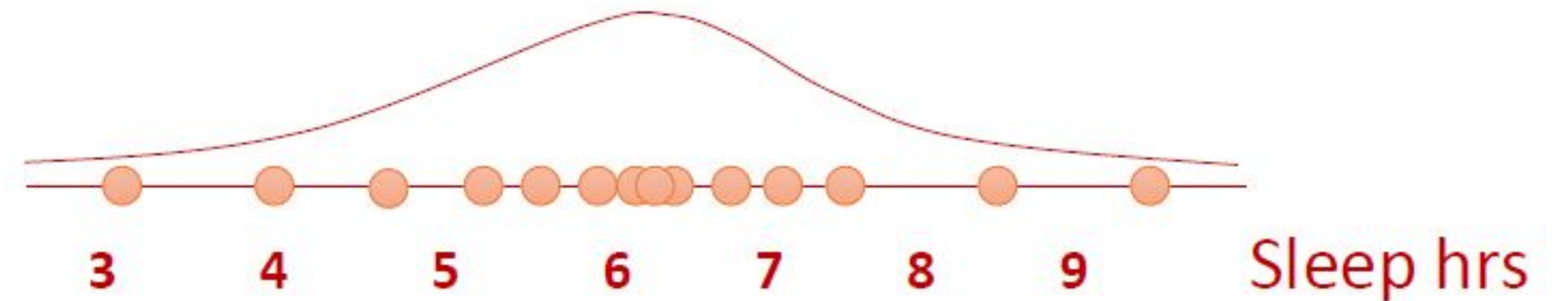
Sleep hours are i.i.d

- independent events
- identically distributed according to Gaussian distribution

Goal: estimate μ , σ

MLE for Mean of Gaussian

Observed data D:



Probability of i.i.d . samples $D=\{x_1, \dots, x_N\}$

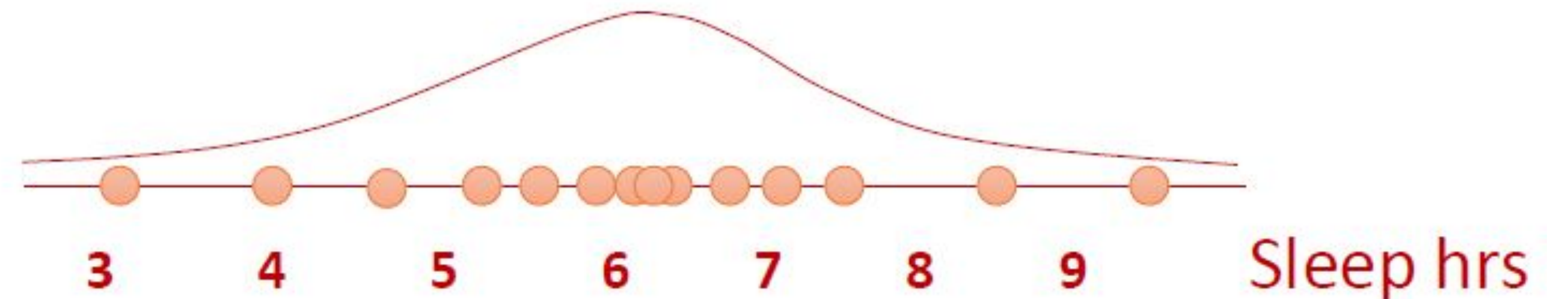
$$P(D \mid \mu, \sigma) = \left(\frac{1}{\sigma \sqrt{2\pi}} \right)^N \prod_{i=1}^N e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

Log likelihood of data

$$\begin{aligned} \ln P(D \mid \mu, \sigma) &= \ln \left[\left(\frac{1}{\sigma \sqrt{2\pi}} \right)^N \prod_{i=1}^N e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right] \\ &= -N \ln(\sigma \sqrt{2\pi}) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} \end{aligned}$$

MLE for Mean of Gaussian

Observed data D:



Probability of i.i.d . samples $D=\{x_1, \dots, x_N\}$

$$P(D | \mu, \sigma) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^N \prod_{i=1}^N e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

Log likelihood

$$\begin{aligned} \ln P(D | \mu, \sigma) &= \ln \left[\left(\frac{1}{\sigma\sqrt{2\pi}} \right)^N \prod_{i=1}^N e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right] \\ &= -N \ln(\sigma\sqrt{2\pi}) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} \end{aligned}$$

Set Derivative = 0

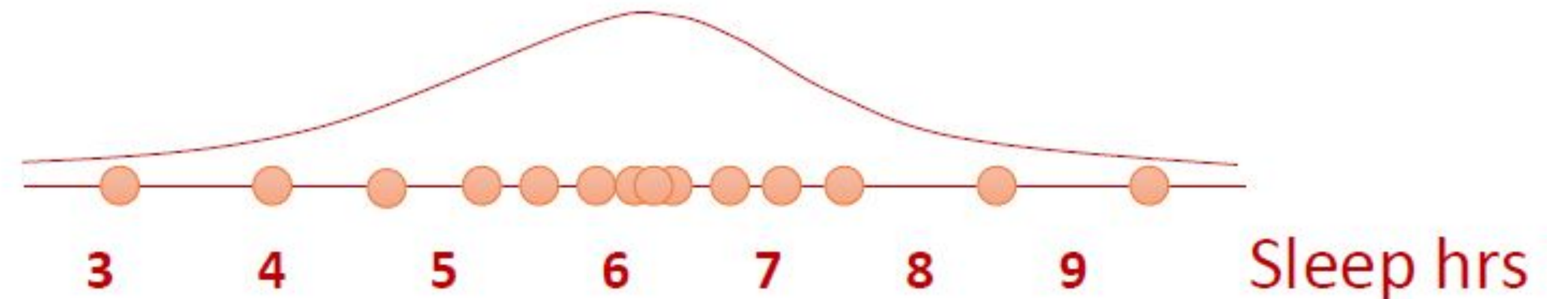
$$\begin{aligned} \frac{d}{d\mu} \ln P(D | \mu, \sigma) &= - \sum_{i=1}^N \frac{d}{d\mu} \frac{(x_i - \mu)^2}{2\sigma^2} \\ &= 2 \sum_{i=1}^N \frac{(x_i - \mu)}{2\sigma^2} = 0 \end{aligned}$$

$$\hat{\mu}_{\text{MLE}} = \frac{\sum_{i=1}^N x_i}{N}$$

$$\sum_{i=1}^N (x_i - \mu) = 0$$

MLE for Variance of Gaussian

Observed data D:



Probability of i.i.d . samples $D=\{x_1, \dots, x_N\}$

$$P(D | \mu, \sigma) = \left(\frac{1}{\sigma\sqrt{2\pi}} \right)^N \prod_{i=1}^N e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

Log likelihood

$$\begin{aligned} \ln P(D | \mu, \sigma) &= \ln \left[\left(\frac{1}{\sigma\sqrt{2\pi}} \right)^N \prod_{i=1}^N e^{-\frac{(x_i - \mu)^2}{2\sigma^2}} \right] \\ &= -N \ln(\sigma\sqrt{2\pi}) - \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^2} \end{aligned}$$

Set Derivative = 0

$$\begin{aligned} \frac{d}{d\sigma} \ln P(D | \mu, \sigma) &= -N \frac{d}{d\sigma} \ln(\sigma\sqrt{2\pi}) - \sum_{i=1}^N \frac{d}{d\sigma} \frac{(x_i - \mu)^2}{2\sigma^2} \\ &= -\frac{N}{\sigma} + 2 \sum_{i=1}^N \frac{(x_i - \mu)^2}{2\sigma^3} \end{aligned}$$

$$\hat{\sigma}_{\text{MLE}} = \frac{\sum_i (x_i - \hat{\mu})^2}{N}$$

Learning Gaussian Parameters

MLE

$$\hat{\mu}_{\text{MLE}} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$\hat{\sigma}_{\text{MLE}}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu}_{\text{MLE}})^2$$

Bayesian learning/estimation is also possible.

Conjugate priors:

- Mean: Gaussian prior
- Variance: Wishart distribution

Summary

- MLE estimates parameters using the observed data.
- MAP combines the likelihood with prior knowledge